

Professional Summary

Advanced C++ Developer with extensive experience in designing, developing, and optimizing high-performance applications. Proficient in object-oriented programming, multi-threading, and software architecture, with a strong background in system programming, embedded systems, and game development. Passionate about writing efficient, scalable, and maintainable code.

Technical Skills

- **Programming Languages:** C++, Python, C, JavaScript
- **Frameworks & Libraries:** STL, Boost, Qt, OpenGL, DirectX
- **Development Tools:** Visual Studio, GCC, Clang, CMake, Git, Docker
- **Operating Systems:** Windows, Linux, macOS
- **Databases:** MySQL, PostgreSQL, SQLite
- **Software Design:** OOP, Design Patterns, Data Structures & Algorithms
- **Networking & Multithreading:** TCP/IP, UDP, Concurrent Programming

Work Experience

Senior C++ Developer *Company Name — Location — 2018 - Present*

- Developed high-performance applications utilizing advanced C++ features and STL.
- Implemented multithreading and concurrent processing to optimize system performance.
- Designed and maintained scalable architectures for enterprise-level software.
- Debugged and optimized code to ensure stability and efficiency.
- Collaborated with cross-functional teams to deliver high-quality software solutions.

C++ Software Engineer *Company Name — Location — 2014 - 2018*

- Developed software for embedded systems and real-time applications.
- Integrated APIs and third-party libraries for enhanced functionality.
- Conducted unit testing and debugging for improved code reliability.
- Refactored legacy code to modern C++ standards, enhancing performance.
- Worked on network programming and low-level system programming.

Education

Bachelor of Science in Computer Science *University Name* — 2010 - 2014

Certifications & Training

- Advanced C++ Programming Certification – *Institution Name* — 2016
- Multithreading & Concurrency in C++ – *Institution Name* — 2017
- Design Patterns in C++ – *Institution Name* — 2018

Projects

- **High-Performance Trading System** – Developed a low-latency trading platform with multithreading and optimized algorithms.
- **3D Game Engine** – Created a custom game engine using OpenGL and DirectX with physics and AI integration.
- **Embedded Systems Controller** – Built firmware and software solutions for IoT devices with C++ and RTOS.

Soft Skills

- Problem-Solving
- Team Collaboration
- Communication
- Analytical Thinking
- Attention to Detail

Languages

- English: Fluent
- Urdu: Native